

GLightOne Base

Technical Documentation - English

Compact technical manual

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1. Introduction

GLightOne is an automotive lighting-control platform designed for 12 V systems. It focuses on LED sequences, coordination between modules and intelligent response to real vehicle signals.

The product was developed as a compact intelligent unit capable of controlling lighting patterns, DRL behavior, turn indication, position, reverse and brake responses while keeping an adaptable architecture for custom automotive installations.

GLightOne Base operates autonomously through physical inputs and internal logic. TFT Lite may be added as an optional visualization and control layer, but it does not replace the base function of the module.

In paired or Full Kit installations, GLightOne can coordinate multiple modules through a Master / Slave architecture and a dedicated SYNC line. The goal is to combine intelligence, design, control and safety in a compact module that respects OEM signals such as STOP, REV, DIR and POS.

2. Product Description

GLightOne Base converts real vehicle signals into coordinated lighting behavior without replacing the original low-beam or high-beam functions of the vehicle.

Each module can control multiple independent LED channels through PWM for sequences, progressive turn-on, dimming, welcome effects, DRL, auxiliary lights, side markers, emergency lighting, underglow or other LED groups installed by the user.

The module reads automotive signals such as POS, DIR, DRL_SW, REV and STOP and turns them into programmed visual responses. The behavior depends on internal logic, mode priority and the installed firmware version.

POS is used as a vehicle state reference, DRL is used for daytime or presence lighting, DIR activates directional response, REV can trigger reverse-related behavior, and STOP provides brake-related response.

The main physical controls are DRL_SW and MODE. DRL_SW manages the DRL function. MODE changes modes and internal behavior. The module receiving these controls normally acts as the Master in a synchronized installation.

All modules use the same base hardware and firmware. There is no different PCB for Master or Slave; the role is defined by wiring and installation. SYNC coordinates the visual behavior between compatible modules.

TFT Lite can be added as an optional interface, while GLightOne Base continues to operate through its physical inputs, LED outputs, MODE, DRL_SW and internal firmware logic.

3. Main Features

- Multiple independent PWM LED channels for sequences, dimming, progressive effects and customized visual behavior.
- 12 V automotive integration with protected power, correct polarity and reliable GND required.
- Reading of real vehicle signals: POS, DIR, DRL_SW, REV and STOP according to the installed configuration.
- Priority logic that keeps critical or functional vehicle signals clear over decorative effects.
- Integrated DRL control through DRL_SW and internal firmware behavior.
- MODE button for mode selection, sequence control, auxiliary functions and configuration according to firmware.
- Master / SYNC architecture for paired or multi-module installations. Full Kit configurations may coordinate up to four modules depending on the kit.
- Optional TFT Lite interface for visualization and technical control without making the base module dependent on the screen.
- Expandable GLight platform prepared for future versions, special configurations and approved integrations.

4. Kit Contents

The kit contents may vary depending on product version, vehicle configuration and installation type. GLightOne Base can be used in paired systems or wider synchronized installations.

GLightOne Base Pair

- Two GLightOne Base modules using the same hardware and base firmware.
- One main control point: the module where DRL_SW and MODE are connected acts as the Master.
- DRL_SW control, MODE button, input connections for vehicle signals, SYNC line and independent PWM LED outputs according to the kit.

GLightOne Full Kit

- Up to four GLightOne Base modules for larger vehicle zones such as front, rear, side areas, auxiliary lights, markers, underglow or visual effects.
- One module defined as Master and the remaining modules working as synchronized units through SYNC.

Optional or project-dependent elements

- TFT Lite interface, custom wiring, harness extensions, external connectors in the vehicle harness and mounting elements may vary by order.

Items not necessarily included by default

- External LED strips, complete headlights, load resistors, relays, external fuses, full vehicle wiring, installation tools, programmers, lab power supplies or OEM vehicle components, unless specifically stated.

5. Product Identification

GLightOne Base may be identified by product name, hardware revision, kit configuration and external security label. The commercial serial number is not necessarily printed directly on the PCB.

All modules use a common base PCB. Physical PCB markings identify the model, hardware revision or general board design, but not necessarily each individual commercial unit.

The commercial serial number is assigned through an external security or identification label. This label may be placed on packaging, protective bag, documentation, warranty card or another area defined by GLight.

The Master is not identified by a different serial number or different board. The Master is the module where DRL_SW and MODE are connected during installation.

Special units such as First Edition, internal reference units or validation units may be differentiated by serial number, documentation, security label or GLight internal record.

6. Electrical Specifications

GLightOne Base is designed for 12 V automotive systems. The final electrical limits may vary by hardware revision, kit configuration and connected LED load.

6.1 Main power

The module must be powered from a 12 V line with correct polarity and a reliable GND. A suitable external fuse is recommended close to the power source. 24 V systems require external adaptation and are not a direct-use condition.

6.2 Internal regulation

The module uses internal regulated rails for logic and auxiliary circuits. These internal rails must not be used as general accessory power outputs unless specifically documented by GLight.

6.3 Inputs

POS, DIR, DRL_SW, REV and STOP are intended for compatible automotive 12 V signals according to the defined installation. SYNC is a system signal for compatible modules. MODE is a user input activated by connecting the button to GND.

6.4 Physical controls

DRL_SW is activated by 12 V. MODE is activated by GND. This difference is essential: DRL_SW = 12 V input; MODE = button to GND.

6.5 Outputs

LED outputs, DRL, OUT STOP, OUT REV, EXP and AUX / EMERGENCY are low-side outputs. They control the negative / GND side of the LED load and do not provide positive 12 V power.

6.6 Loads

The positive side of each LED load must receive the correct power from the installation. The negative side of that load connects to the corresponding GLightOne output. Loads must be compatible with the driver, system and installed configuration.

6.7 SYNC

SYNC coordinates compatible GLightOne modules. It requires common GND between modules. SYNC must never receive +12 V and must not be connected to OEM high-current signals or external loads.

6.8 Important note

Before powering the system, verify +12V, GND, fuse, polarity, LED load wiring, DRL_SW as 12 V input and MODE as button to GND.

7. Connection Pads and Pinout

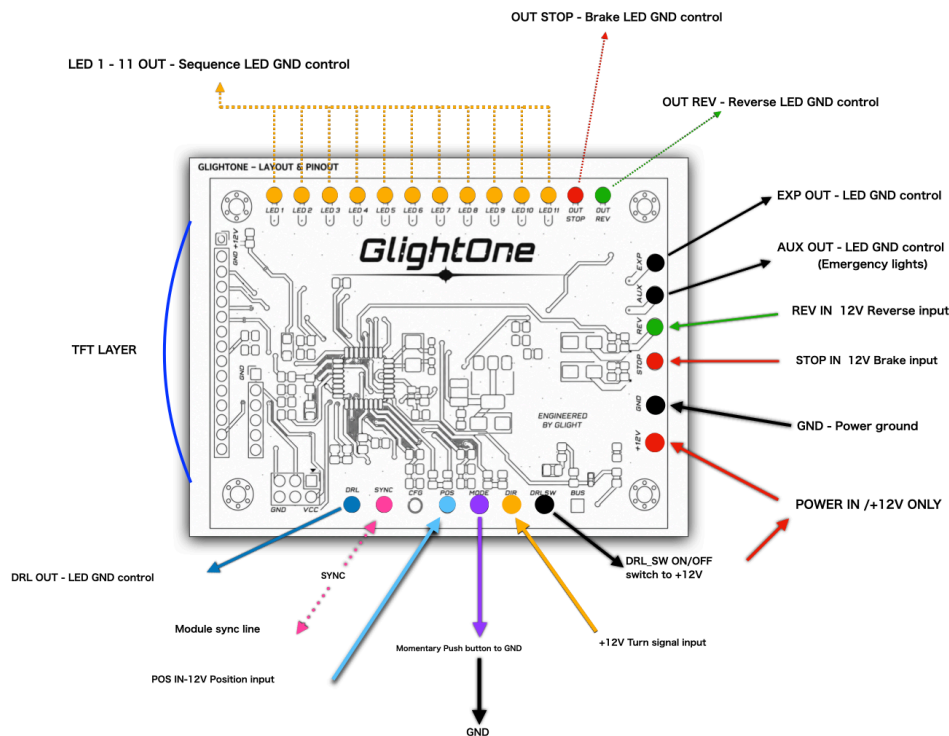


Figure 1 — GLightOne Base connection pad and pinout map

Note: LED outputs are low-side / GND control. They do not provide +12V.

MODE requires a momentary push button to GND. DRL_SW uses an ON/OFF switch to +12V. GLightOne Base uses connection pads identified by silkscreen marking. This section must be used together with the physical board diagram of the specific hardware revision. No image is included in this compact translated file.

7.1 Main power

+12V receives the positive 12 V supply for the module. GND is the main ground reference. In multi-module installations, all modules must share common GND.

7.2 Vehicle signal inputs

POS detects position or lighting state. DIR detects turn signal. DRL_SW receives 12 V for DRL control. REV detects reverse. STOP detects brake. These pads must not be used as outputs or power pads.

7.3 MODE

MODE is the main user button input. It is active to GND: one side of the button goes to MODE and the other side goes to GND. MODE must not receive +12 V.

7.4 SYNC

SYNC is a shared synchronization line between compatible GLightOne modules. Do not solder many wires directly onto one SYNC pad; use an external splice, connector, terminal, harness or distribution point outside the PCB when several modules are involved.

7.5 Main LED outputs

LED 1 to LED 11 are PWM outputs for sequences, chasers, dimming and custom behavior. They control the negative side of the LED load. The LED positive must be powered separately by the installation.

7.6 Special outputs

DRL, OUT STOP, OUT REV, EXP and AUX / EMERGENCY are functional outputs and also control the negative / GND side of their loads. They must not be treated as positive 12 V outputs.

7.7 Reserved or system pads

CFG and BUS are reserved or system-related pads depending on hardware and firmware version. They must not be connected to +12 V, GND, OEM signals or external devices without official GLight documentation. TFT Layer is reserved for compatible TFT Lite integration.

7.8 Quick summary

+12V = positive supply. GND = main ground. POS/DIR/DRL_SW/REV/STOP = 12 V signal inputs. MODE = button to GND. SYNC = synchronization line, not power. LED outputs and special outputs = negative / GND control.

8. Basic Installation

Basic installation must follow the official pinout, electrical architecture and installation configuration. The vehicle must be off and power disconnected while wiring or soldering.

- Prepare the installation: check kit version, module positions, LED zones, signal points, wire routing and which module will act as Master.
- Define the system: a paired system normally uses two modules; a Full Kit may use up to four synchronized modules depending on the project.
- Define the Master: the module where DRL_SW and MODE are connected acts as the main control point. Only one main control point should exist in a standard installation.
- Connect +12V and GND: use protected 12 V power, reliable ground and common GND between modules.
- Connect LED loads: LED positive to the correct supply, LED negative to the corresponding GLightOne output.
- Connect controls: DRL_SW as 12 V input and MODE as button to GND.
- Connect OEM signals only after measuring and confirming POS, DIR, REV and STOP as required by the installation.
- Connect SYNC between compatible modules only, with common GND and no +12 V on the SYNC line.
- Perform an initial test before closing panels: power, MODE, DRL_SW, POS, DIR, REV, STOP, SYNC and LED loads according to the actual installation.

Before driving, confirm that STOP and DIR remain clear, visible and stable, and that no OEM critical function is compromised.

9. General Operation

After receiving 12 V power, GLightOne Base initializes its internal logic and begins reading connected signals, physical controls and synchronization state.

The module operates autonomously using its firmware, physical inputs, PWM LED outputs and SYNC line when multiple modules are installed.

In normal conditions, the active visual mode, DRL behavior, presence lighting or sequence runs while the firmware continues monitoring priority signals.

STOP has safety priority as a brake-related signal. DIR has signaling priority and must remain clear and dominant. REV may activate reverse-related response. POS may modify lighting state. DRL can be dimmed, paused or modified when a priority signal requires it.

MODE changes functions according to the firmware and system state. DRL_SW manages DRL behavior. In multi-module systems, the Master sends coordination through SYNC and synchronized modules follow the expected behavior.

TFT Lite, when present, is an optional interface. GLightOne Base does not require the screen to operate.

Correct operation depends on protected power, reliable GND, compatible loads, correct signals and final validation before vehicle use.

10. Lighting Modes

A lighting mode is a programmed behavior that defines how outputs turn on, turn off, dim or sequence. Modes apply when no priority signal requires a different response.

10.1 MODE with DIR active

When DIR is active, MODE is dedicated to the directional function. One short press changes the directional sequence. A long press changes sequence speed. Double, triple or quadruple presses are ignored in this condition to avoid accidental auxiliary changes during turn indication.

10.2 MODE with DIR inactive

When DIR is not active, MODE can control auxiliary visual functions depending on firmware: one press changes DRL mode, two presses change EXP mode, three presses change Emergency mode including OFF cycling, four presses change STOP mode, and a long press changes Welcome mode.

10.3 Visual functions

Sequence modes, Flow, Welcome, EXP and AUX / EMERGENCY can create progressive effects, smooth transitions, presentation sequences, warning behavior or auxiliary lighting according to firmware and installation.

10.4 Relationship with OEM signals

POS, DRL_SW, DIR, STOP and REV may modify the active visual mode. When STOP, DIR or REV is active, the system may pause, replace, dim or redirect decorative effects to keep the vehicle signal clear.

10.5 Synchronized modes

In multi-module installations, modes can be coordinated through SYNC so the vehicle behaves as a single visual system rather than isolated modules.

10.6 Important note

MODE controls visual behavior, but it must never make STOP, DIR or REV confusing, weak or delayed.

11. DRL / POS Function

DRL / POS allows GLightOne Base to manage daytime lighting, presence lighting and position-related behavior through real vehicle signals and internal logic.

DRL is an active visual function. POS is a reference for vehicle lighting state. They are related but not the same.

DRL_SW is the dedicated physical control for DRL and is activated by 12 V. When DRL_SW is active, the module may enable or manage the DRL behavior configured in firmware.

When POS is active, the system may change brightness, presence behavior, night behavior or another state defined by firmware. POS does not necessarily replace the OEM position-light function.

DRL must not block priority signals. DIR must remain visible and dominant, STOP must remain clear, and REV may modify the visual response when installed.

In synchronized systems, DRL can be coordinated by SYNC while all modules share common GND.

Use DRL / POS to create clean presence lighting without compromising turn, brake or reverse signals.

12. DIR / Turn Signal Function

DIR allows GLightOne Base to respond to the vehicle turn signal and execute a directional visual behavior.

DIR is a priority signal. Its purpose is to keep the turn indication clear and visible over DRL, decorative effects or auxiliary modes.

When DIR is active, the module may run a directional sequence, sweep, flow pattern or firmware-defined response. DRL, EXP, AUX or other visual effects may be dimmed, paused or replaced temporarily.

With DIR active, MODE changes the directional sequence with a short press and changes sequence speed with a long press. Multiple auxiliary presses are not used during DIR active state.

In paired or synchronized installations, the installer must verify left/right assignment, sequence direction, SYNC behavior and return to normal state. SYNC must not create a wrong or confusing turn indication.

Before driving, verify that DIR activates the correct side, remains stable during blinking and is not hidden by any visual mode.

13. REV / Reverse Function

REV allows GLightOne Base to respond to the vehicle reverse signal and activate an auxiliary visual response associated with backing up.

REV is a functional priority signal. It may activate OUT REV, rear auxiliary lighting or another firmware-defined reverse behavior. It does not replace the original vehicle reverse function.

OUT REV is a low-side output. It controls the negative / GND side of the connected load and does not provide positive power.

When REV is active, DRL or other effects may be modified to avoid visual confusion. If REV coincides with DIR, the turn indication must remain understandable. If REV coincides with STOP, brake clarity must be preserved.

In paired or synchronized installations, REV behavior must remain stable and coherent between modules. SYNC must not create ambiguous rear indications.

If REV is not part of the installation, it may be omitted and must not be considered a module fault.

14. STOP Function - Safety First

STOP allows GLightOne Base to respond to the vehicle brake signal with a clear, immediate and stable visual response.

STOP is a critical OEM signal. Inside GLightOne it must be treated as a visual safety function, not as a decorative animation.

When STOP is active, the system may modify DRL, sequences or auxiliary effects so that the brake indication remains recognizable.

MODE must not make the brake response confusing.

OUT STOP is a low-side output dedicated to a brake-related LED load when the installation uses that function. It controls the negative / GND side of the load and does not provide positive 12 V.

The STOP response should be simple, clear and fast. It should not depend on long animations or slow transitions that delay recognition.

If STOP coincides with DIR or REV, the final response must remain safe and understandable. Neither DIR nor REV should hide or reduce brake clarity.

STOP must be tested before vehicle use. Verify the input, OUT STOP if installed, interaction with DRL, DIR, REV, MODE and SYNC, and correct return to normal behavior.

15. Master / Slave Synchronization

GLightOne Base can operate with more than one module using a Master / Slave architecture defined by installation.

All modules use the same base hardware and main firmware. There is no separate Master PCB or separate Slave PCB in the standard system.

The Master is the module receiving the main controls, normally DRL_SW and MODE, plus the required signals for that installation. The other modules follow the synchronized behavior through SYNC.

SYNC is a signal line, not power. It must not receive +12 V and must not be connected to power lines, LED outputs, high-current OEM signals or external loads.

Only one main control point should exist in a standard installation. Duplicated MODE or DRL_SW controls can create contradictory behavior unless specifically designed and validated by GLight.

In paired or Full Kit systems, every module must control the correct vehicle zone. SYNC coordinates behavior but does not correct physically inverted wiring or wrong side assignment.

Before driving, verify Master role, synchronized module response, common GND, SYNC continuity, MODE, DRL_SW, POS, DIR, REV, STOP and return to normal state.

16. MODE Button

MODE is the main physical button for changing functions and visual modes from the system control point.

MODE is active to GND. One side of the button connects to MODE and the other side connects to GND. MODE must not receive +12 V, must not be connected to LED outputs and must not be connected to power lines.

With DIR active, MODE is limited to directional functions: short press changes sequence, long press changes sequence speed, and multiple auxiliary presses are ignored.

With DIR inactive, MODE can control normal auxiliary functions depending on firmware: short press for DRL mode, two presses for EXP / Angel mode, three presses for Emergency mode and OFF cycle, four presses for STOP mode, and long press for Welcome mode.

MODE must normally be connected to the Master in synchronized installations. Duplicating MODE on several modules is not recommended unless the configuration is specifically defined and validated.

After installation, verify every MODE action, its relation with DIR active/inactive state, and that it does not compromise POS, DIR, REV or STOP behavior.

17. Optional TFT Lite Interface

TFT Lite is an optional visualization and technical control layer for compatible GLightOne Base versions.

GLightOne Base remains an autonomous module. Its main operation is handled by physical inputs, OEM signals, MODE, LED outputs, SYNC and internal firmware logic.

TFT Lite may display system states, active modes, selected functions or touch controls depending on firmware. It can help during installation, validation, demonstration or advanced use.

TFT Lite must connect only to the pads or area identified as TFT Layer, following compatible orientation, wiring and GLight instructions. It must not be connected to unrelated pads or external devices.

In multi-module systems, TFT Lite is normally connected to the Master or main control point. It may coexist with MODE and with OEM signals; the firmware keeps priority signal response under control.

The screen must be installed safely, visible and protected, without blocking driver view, original controls, airbags or safety elements. It must not distract the driver.

18. Firmware and Version Control

GLightOne Base uses dedicated firmware developed by GLight to control inputs, LED outputs, visual modes, synchronization, internal memory and responses to vehicle signals.

The standard product is delivered with firmware installed, tested, validated, protected and registered according to GLight internal process. The firmware is not supplied as open source code or as a freely editable file.

Commercial units may include protection against unauthorized reading, copying, extraction, modification or replacement. Unauthorized intervention may cause configuration loss, module lockout, incorrect behavior, service requirement or warranty loss.

GLight may develop special firmware versions for volume orders, specific projects, commercial applications or approved integrations. These are controlled and validated product variants, not open end-user configuration.

Direct programming by a customer or installer is not the standard flow. Any programming, update, recovery or service process must be specifically authorized by GLight.

Firmware versions may vary by batch, kit, production date, configuration or approved special version. Serial number, security seal, batch and firmware reference help support, warranty and traceability without exposing protected firmware.

The Master / Slave role is defined by installation, not by different firmware. Programming or service interfaces, if present, are considered production, validation or authorized service tools.

19. Validation Checklist

After installation, configuration or authorized service, the system must be validated before delivery or public-road use. Validate only the functions actually installed and configured.

- Inspect the module, soldered pads, wire protection, mounting, isolation, polarity and possible shorts before powering.
- Verify +12V, GND, external fuse, stable power and common GND between modules.
- Verify each connected LED output: LED positive to the correct supply and LED negative to the GLightOne output. Outputs must not be treated as positive power.
- Check DRL and DRL_SW if installed. Confirm behavior with POS, DIR, REV or STOP if those signals are connected.
- Check POS and DIR if connected. DIR must activate the correct side and remain clear and dominant.
- Check REV and OUT REV only if reverse is part of the installation. If not installed, this test does not apply.
- Check STOP and OUT STOP only if brake response is part of the installation. If installed, STOP must be clear, stable and visible.
- Check MODE as button to GND and verify its behavior with DIR active and inactive.
- Check EXP and AUX only if included in the configuration.
- Check SYNC only in multi-module installations. Confirm common GND, correct module zones and coherent response.
- Check TFT Lite only if included. The screen, touch and commands must behave correctly while physical signals remain valid.
- Verify return to normal state after each signal is removed. The system must not remain locked in a previous condition.

No installation should be considered complete until all connected functions are physically verified on the real LED loads.

20. Recommended Installation

Install GLightOne Base in a protected, dry, secure and accessible area. Avoid water, constant condensation, extreme heat, oil, chemicals, severe vibration, impacts and moving parts.

Use the silkscreen as the main connection reference. Every pad must be used only for its indicated function. Do not reuse unused pads for improvised signals or power.

Solder wires cleanly and firmly to the pads. Avoid solder bridges, cold joints, conductive residue and mechanical stress on pads. Use strain relief and wiring protection whenever necessary.

Power must respect polarity: +12V to the +12V pad and reliable GND to the GND pad. Use an appropriate external fuse close to the source and choose wire gauge according to load.

Keep wiring organized and protected from heat, friction, sharp edges and movement. Separate power lines, LED outputs and control signals when possible.

Connect POS, DIR, DRL_SW, REV and STOP only to verified compatible vehicle signals. Measure unknown OEM points before connection.

MODE must be installed as a button to GND. SYNC must be used only as synchronization between compatible modules and must never receive +12 V.

LED outputs, OUT STOP, OUT REV, EXP and AUX control the negative / GND side of their loads. They must not be used as positive supply or signal inputs.

TFT Lite, if installed, must connect only to the TFT Layer area and be placed safely without obstructing driving or safety systems.

External connectors may be used in the vehicle harness outside the module, but they do not change the electrical function of each soldered pad.

Before final closure of panels or covers, perform a functional test of power, GND, connected inputs, outputs, MODE, SYNC, TFT Lite and all installed functions.

21. Common Errors and Basic Diagnosis

Diagnosis should start with installation, power, ground, pads, soldering, signals and LED loads before assuming an internal module defect.

- Module does not power on: check +12V, GND, polarity, fuse, wire gauge, voltage drop and solder joints.

- Intermittent behavior: check weak ground, cold solder, wire movement, vibration, loose splices, humidity, corrosion or contact between pads.
- Wrong pad: verify the silkscreen name and function before powering. A nearby or unused pad is not interchangeable.
- LED output used as positive: GLightOne outputs control GND. LED positive must come from the correct supply.
- Incorrect load: excessive current or incompatible LED architecture can cause resets, heating or low brightness.
- MODE does not respond: verify MODE to GND, button wiring, soldering and that MODE is connected to the Master.
- MODE triggers unexpected functions: check DIR state, multiple presses, button bounce and firmware behavior.
- DRL, POS or DIR issues: verify actual signal voltage, correct pad, left/right assignment and interaction with other functions.
- REV or STOP issues: diagnose only if those functions are installed. OUT REV and OUT STOP are outputs, not signal inputs.
- EXP or AUX issues: verify that they are enabled in the configuration and that loads are wired as low-side outputs.
- SYNC issues: verify common GND, continuity, correct module roles and no +12 V on SYNC.
- TFT Lite issues: check TFT Layer connection, compatibility, orientation, firmware and touch calibration if applicable.
- Memory or strange resets: check supply stability, GND, load current, floating unused inputs, humidity and overloading.

Never connect programmers, reading tools or unauthorized service interfaces without GLight instruction. Stop testing immediately if there is smoke, burning smell, abnormal heat, repeated fuse blowing, severe voltage drop or visible damage.

22. Warranty, Support and Limitations

The warranty and support of GLightOne Base are intended to protect the user against verifiable product defects while protecting GLight from damage caused by incorrect installation, unauthorized handling, use outside specification or system intervention.

The warranty may cover manufacturing defects, verifiable internal failures or product-attributable problems under normal use and correct installation. GLight may evaluate each case before approving repair, replacement or warranty support.

For support, GLight may request serial number, security seal condition, model, kit type, purchase or delivery date, photos, video, +12V and GND measurements, connected load type and description of the used signals.

The warranty does not cover incorrect installation, bad soldering, lifted pads, reverse polarity, external short circuits, overload, moisture, water, oil, chemicals, extreme heat, vibration, impacts, physical modification, unauthorized tools or use outside specification.

It also does not cover damage caused by applying +12 V or wrong voltage to MODE, SYNC, LED outputs, OUT STOP, OUT REV, TFT Layer or any other non-power pad.

External LED loads, vehicle wiring, fuses, splices, external connectors, accessories, third-party installation and incompatible external components are outside the module warranty unless applicable law requires otherwise.

Firmware, internal memory and protected configuration are part of the product. Unauthorized attempts to read, extract, copy, modify, overwrite or replace firmware may limit or void warranty support.

The security seal and serial number support traceability. They must not be removed, altered, covered, forged, transferred or duplicated.

GLightOne Base is part of the GLight technical platform. Hardware, firmware, architecture, functional behavior, documentation, visual identity, configurations, implementation methods and commercial protection elements are GLight intellectual property. Purchase grants product use within its intended purpose, not rights to firmware, source code, design, brand, methods, production tools or platform exploitation.

This warranty applies without prejudice to any mandatory user rights under the applicable laws of the country or region.

23. Safety and Use Warnings

GLightOne Base is a specialized automotive electronic module. It must be installed, powered, used and validated according to technical documentation, silkscreen marking, connection pads and purchased configuration.

Power must be disconnected during installation, soldering, pad inspection, wiring modification or load connection. Do not solder, cut or move wires while the system is powered.

+12V must only be applied to the +12V pad. GND must be reliable. Do not reverse polarity and do not apply +12 V to signal or output pads.

Each pad must be used only as indicated. MODE, SYNC, POS, DIR, DRL_SW, REV, STOP and TFT Layer must not be used as improvised power points.

LED outputs and special outputs control the negative / GND side of loads. Unknown, overloaded, short-circuited or out-of-spec loads must not be connected.

MODE must be a button or control to GND and must never receive +12 V. SYNC must be used only between compatible modules, with common GND, and must never receive +12 V.

REV, STOP, EXP, AUX, SYNC and TFT Lite are optional depending on the defined installation. If they are not used, do not connect them in improvised ways.

TFT Lite, if installed, must not block driving, driver visibility, original controls, airbags or safety elements, and must not distract the driver.

Protect the module from water, excessive humidity, corrosion, mud, oil, chemicals, extreme heat, severe vibration, impacts and moving parts. Use strain relief on soldered wires.

Do not interfere with airbags, brakes, steering, ECU, sensors, ABS, fuel lines or other critical vehicle systems. Measure and confirm OEM signals before connection.

Visual functions must comply with local laws. Some colors, intensities, sequences or special modes may not be allowed on public roads.

Do not continue using the system if there is smoke, burning smell, abnormal heat, repeated fuse blowing, strong voltage drop, severe erratic behavior, damaged pads or shorted LED loads.

Do not connect programmers, reading tools or unauthorized external interfaces. Firmware is a protected part of the product.

The system should be installed by a person with basic knowledge of automotive electricity, soldering, polarity, signal measurement and wiring protection. Test all connected functions before driving.

24. Final Specifications and Technical Summary

GLightOne Base is an automotive electronic visual-control module designed to manage LED functions, vehicle signals, internal modes, module synchronization and visual configurations according to the installed firmware version.

Commercial name: GLightOne Base. Product type: automotive electronic LED visual-control module. Platform: GLight. Main application: control of automotive visual functions, LED modules, DRL, sequences, OEM signals, synchronization and auxiliary functions according to configuration.

Architecture: autonomous module with hardware, firmware, connection pads, LED outputs, signal inputs, MODE control, SYNC synchronization and internal memory according to version. Master / Slave role is defined by installation and wiring.

Power: 12 V nominal automotive supply. +12V receives positive power and GND must connect to reliable ground. Stable supply and GND are essential.

Physical connection: soldered pads identified by silkscreen marking. The module is not an integrated removable-connector product. External connectors may be used in the vehicle harness outside the module.

Inputs may include POS, DIR, DRL_SW, REV, STOP, MODE, SYNC and optional signals according to version and configuration. Not all inputs must be connected in every installation.

Outputs may include main LEDs, DRL, OUT STOP, OUT REV, EXP, AUX, Emergency and special visual functions. Outputs control negative / GND and must not be used as positive supply.

Functions may include DRL, LED sequences, DIR mode, POS response, REV response, STOP response, Welcome Mode, Emergency, EXP, AUX, mode memory, synchronization and MODE control.

REV and STOP are optional according to installation. EXP and AUX are optional according to configuration. TFT Lite is optional in compatible versions and does not replace the base module logic.

Firmware is developed by GLight, installed, validated, protected and version controlled. Internal memory may store selected modes depending on version. Serial number and security seal support identification and traceability.

Every connected function must be correctly soldered, protected, identified and validated before use. GLightOne Base must be used within specification, documentation, warranty conditions, intellectual-property protection and applicable regulations.